Aritra Majumdar

Heritage Institute of Technology, Kolkata
B.Tech CSE (Data Science)

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GitHub Profile
LinkedIn Profile

Education

•Heritage Institute of Technology, Kolkata *2024 – 2028*
B.Tech in CSE (Data Science) CGPA: 7.16

•Bodhicariya Senior Secondary School *2024*
XII Standard (CBSE), PCM Percentage: 73.5%

•St. Stephens School *2022*
X Standard (ICSE), PCM Percentage: 90.18%

Experience

•Freelancing *April 2025 – Present*
Data Science, Finance & GenAI Developer Remote

- Working on diverse projects involving Data Science, Quantitative Finance, and Generative AI.
- Developing Backend architectures and Blockchain solutions.

Personal Projects

•AI Trading Indicator
Algorithmic Trading & Analysis

- Tools used: Python, MT5, VectorBt, Pandas, Numpy, Scikit-learn
- Developed an indicator for the top 10 most volatile instruments using statistical methods and data modeling.
- Built data pipeline for preprocessing and validation of 20 years of historical data with 10,000 backtested scenarios.

•Viva Voce AI Agent
Educational RAG & NLP Tool

- Tools used: Ollama, Python, JS, CSS, HTML, React, Vector DBs, LLM APIs
- Preparation helper using prompt chaining and RAG architecture; generates contextual questions from PDF inputs.
- Features Voice-to-Text capability with API integration and thematic modes (Normal, Money Heist).

•AI Portfolio Manager
FinTech Application

- Tools used: Python, NewsAPIs, Scikit-learn, Ollama, React, HTML/CSS, Jupyter Notebooks
- Utilizes data gathering from live news, fundamentals, and quarterly results for financial analysis and prediction.
- AI-driven solution managing portfolio weightage dynamically using data analysis and visualization of results.

•Tokenization of Assets (Arbitrum)
Blockchain & RWA

- Tools used: Solidity, Golang, Python, React, HTML/CSS, API Integration
- Platform for tokenizing Real World Assets (Real Estate, Paintings) with market research-driven pricing models.
- Enables low-capital entry into high-value asset classes with Excel-based analytical tools for investors.

•Govt. Hospital Bed Referral System
Healthcare Management

- Tools used: Python, Scikit-learn, Numpy, Pandas, React, HTML/CSS
- Live updating system using data collection and data preparation pipelines for patient bed matrix.
- Reduces administrative burden through data visualization and analytical skills applied to referral optimization.

Independent Quantitative Research

•Regime-Adaptive HFT: Stochastic Control in Non-Stationary Markets *In Progress*
Computational Finance & Market Microstructure

- Researching dynamic risk-aversion (γ) tuning within the Avellaneda-Stoikov framework using prompt refinement techniques for model building and parameter optimization.
- Engineering robust data pipeline with data preprocessing, validation, and maintenance for over 100 aggregate years of L1 tick data (10 instruments $\times$ 10 years).
- Analyzing top-of-book microstructure signals using AI frameworks and statistical methods for data modeling and visualization of results to mitigate adverse selection.
- Benchmarking adaptive parameterization with AI/ML projects methodology and financial analysis techniques using 2024 Bitcoin (BTC) high-frequency data.

Technical Skills and Interests

Languages: Python, Golang, C, SQL
ML/AI Tools: Scikit-learn, TensorFlow, PyTorch, LLM APIs, Vector DBs, Jupyter Notebooks
Developer Tools: Git, VS Code, MetaEditor5, Microsoft 365 Copilot, Excel
Cloud/Databases: Google Colab, Google Cloud, SQLite3, PostgreSQL
Core Competencies: Quantitative Finance, Prompt Engineering, Data Structures, Market Research
Areas of Interest: High-Frequency Trading, Machine Learning, Generative AI, Blockchain

Achievements

- **Finalist,** Lexicognition AI (Sponsored by Microsoft), Kshitij - IIT Kharagpur *2026*
- **Top 10 Finalist,** Srijan Techfest (Jadavpur University), Main Hackathon (out of 600 teams)
- **WBJEE Rank,** Secured Rank 6101